

Multiple Choice (10 marks)

1. In the atomic nucleus, electrical forces tend to
 - (a) cause particles to become neutral
 - (b) increase the mass of the nucleus
 - (c) cause particles to spin
 - (d) generate heat within the nucleus
 - (e) create magnetic fields
2. A car battery supplies a current I of 50 amp to the starter motor. How much charge passes through the starter in $(1/2)$ min?
 - (a) 1000 coul
 - (b) 1800 coul
 - (c) 3000 coul
 - (d) 1500 coul
 - (e) 2500 coul
3. Electricity and magnetism connect to form
 - (a) mass
 - (b) low-frequency sound
 - (c) nuclear power
 - (d) energy
 - (e) heat
4. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $2 kT$
 - (b) $(7/2) k T$
 - (c) $5 kT$
 - (d) $(1/4) k T$
 - (e) $3 kT$
5. An airplane lands on a carrier deck at 150 mi/hr and is brought to a stop uniformly, by an arresting device, in 500 ft. Find the acceleration and the time required to stop.
 - (a) -45 ft/sec^2 and 4 sec

- (b) -48.4 ft/sec^2 and 4.55 sec
- (c) -40 ft/sec^2 and 6 sec
- (d) -42 ft/sec^2 and 5.5 sec
- (e) -52 ft/sec^2 and 4.8 sec

True / False (10 marks)

Answer True or False

- 6. True or False: To completely ionize helium, the energy required is 79.0 eV, which is the energy to remove both electrons.
- 7. True or False: The principal source of Earth's internal energy is geothermal heat generated by radioactive decay within the Earth's crust.
- 8. True or False: The mass of an atom comes mostly from its protons, and its volume is influenced by its neutrons.
- 9. True or False: The calculated number of proton decays per day from a 62-kg person is approximately 3.5×10^{-8} .
- 9. True or False: A car engine working at 33 horsepower exerts a force of 300 lbs when moving at 45 mi/hr.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Analyze the motion of a stone thrown horizontally from a bridge at 10 m/s. Calculate the time it takes to strike the water 80 m below, considering gravitational effects and neglecting air resistance.
- 12. Analyze the motion of passengers on a carnival ride moving in a horizontal circle of radius 5.0 m, completing a full circle in 4.0 s. Calculate their acceleration and discuss the implications of this value in the context of circular motion.

End of Paper